

A Framework for Robust and Iterative Access Point Localization from Crowdsourced Wi-Fi Data

1. The Crowdsourced Data Landscape: Characteristics and Inherent Challenges

The development of a robust Access Point (AP) localization system founded upon crowdsourced data necessitates a profound understanding of the data itself. The information provided by user devices is not a clean, deterministic signal but rather a collection of measurements, each laden with multiple layers of uncertainty and error. These errors stem from the fundamental physics of radio frequency (RF) propagation, the inherent limitations of consumer-grade location sensors, and the diversity of the devices collecting the data. A failure to correctly model and mitigate these distinct error sources will inevitably lead to a system with poor accuracy and reliability. Therefore, the first step in designing a successful localization pipeline is a rigorous analysis of the characteristics and challenges of the input data stream.

Anatomy of a Crowdsourced Wi-Fi Measurement: Introducing Data Tiers

The foundational data element for this localization process is a single Wi-Fi observation. However, not all observations are created equal. Data collected from an AP to which the device is actively connected is of significantly higher quality than data from a passive scan. This distinction must be a first-class citizen in the data model.

The atomic unit of data for the localization pipeline should therefore be a tuple that includes this contextual information:

(BSSID, lat, lon, alt, accuracy, RSSI, timestamp, connection_status, connection_metadata)

Where:

- **BSSID, lat, lon, alt, accuracy, RSSI, timestamp:** As previously defined.
- **connection_status:** An enumerated type indicating the source of the measurement, e.g., CONNECTED or SCAN. This is the primary indicator of data quality.
- **connection_metadata:** A rich object containing high-fidelity data available only for CONNECTED measurements. This includes, but is not limited to:
 - frequency: The exact operating frequency (e.g., 5660 MHz).
 - channelWidth: The width of the channel (e.g., 80 MHz).
 - linkSpeed: The negotiated connection speed (e.g., 351 Mbps).

- `is80211mcResponder`: A boolean indicating support for Wi-Fi RTT.
- `capabilities`: A string detailing security and protocol features.

This tiered data model is fundamental. CONNECTED data represents a premium, high-confidence signal, while SCAN data provides broader but less reliable environmental context. The entire pipeline must be designed to leverage this distinction.

The Uncertainty Principle in Crowdsourced Location: Analyzing GPS/GNSS Error

The most significant challenge in using crowdsourced data is the unreliability of the reported client location (lat, lon). This location, which the system must treat as its reference, is itself an estimate generated by the device's Fused Location Provider, typically relying on a Global Navigation Satellite System (GNSS) like GPS. The quality of this estimate is highly variable and subject to several types of error, most notably multipath propagation in urban and indoor environments, which can introduce significant systematic bias.[4, 5] The accuracy field is therefore a critical piece of metadata for filtering low-quality data.

The Physics of Indoor Signal Propagation: Multipath, Attenuation, and Fading

The second primary source of uncertainty lies in the RSSI measurement. The complex indoor environment distorts the relationship between distance and signal strength through path loss, shadowing, and multipath fading.[6, 8, 10] This ensures that the relationship is not a clean function but a noisy, stochastic one that requires robust statistical modeling.

The Challenge of Device Heterogeneity

A final, critical challenge specific to crowdsourcing is device heterogeneity. Different devices will report different RSSI values even when at the exact same location due to variations in hardware and software.[11, 12] This introduces a device-specific systematic bias that must be accounted for in the system design.

2. Handling AP Dynamics: Mobility and Relocation

A foundational assumption in many localization systems is that all access points are static. In a real-world crowdsourced environment, this assumption is frequently violated and presents a critical threat to the integrity of the location database. The system must therefore incorporate a dedicated module to classify APs and detect changes in their state over time.

2.1 Detecting and Filtering Mobile Wi-Fi Hotspots

Mobile hotspots must be proactively identified and excluded from the core localization process. This is achieved by analyzing the long-term characteristics of the data collected for each BSSID.

- **Spatial Distribution Analysis:** A stationary AP will have its associated location reports concentrated in a geographically tight cluster. A mobile hotspot's reports will be spread over a wide area. A high standard deviation of the coordinates is a strong indicator of mobility.
- **Temporal Neighbor Analysis:** A stationary AP is consistently observed alongside a stable set of neighboring APs. A mobile hotspot's neighbors change constantly.
- **SSID and BSSID Heuristics:**
 - **SSID Blacklisting:** Maintain a blacklist of common hotspot patterns ("AndroidAP," "iPhone," etc.).
 - **OUI Lookup:** Use the BSSID to identify the manufacturer. BSSIDs from mobile phone manufacturers are likely hotspots.
 - **connection_metadata:** Fields like `isPasspointNetwork: false` and capabilities containing WPS are strong indicators of consumer-grade, residential APs, which are less likely to be mobile than an AP flagged as `isEphemeral` (if available).

2.2 Detecting the Relocation of Stationary APs

When a stationary AP is moved, all historical data becomes invalid. The system must detect this change to avoid averaging old and new location clusters.

- **Change-Point Detection:** Treat the sequence of location reports as a time series and apply a change-point detection algorithm (e.g., monitoring Mahalanobis distance or using a CUSUM chart).
- **Bi-modal Cluster Detection:** Periodically run a clustering algorithm (e.g., DBSCAN) on the location data. The emergence of two distinct, geographically separate clusters is a strong sign of a move.

2.3 Data Invalidation and State Reset

Once a relocation has been detected and confirmed, the system must take immediate action:

1. **Invalidate Historical Data:** All measurement records prior to the relocation time are marked as invalid for current localization.
2. **Reset AP State:** The AP's "Data Maturity Model" (Section 4) is reset.
3. **Re-enter Bootstrap Phase:** The AP is treated as newly discovered.

3. Foundational Data Pre-processing and Filtering

Before any attempt at localization can be made, the raw stream of crowdsourced measurements must undergo a rigorous, multi-stage filtering and pre-processing pipeline. This pipeline must now incorporate the `connection_status` as a primary indicator of data quality.

Stage 1: Initial Data Ingestion, Sanity Checking, and Quality Weighting

This first stage acts as a coarse-grained filter, immediately discarding records that are clearly invalid and applying an initial quality weighting based on the source of the data.

- **Sanity Checks:** Discard records with missing coordinates, invalid RSSI values, or very high accuracy values (e.g., $> 150\text{m}$).
- **Connection Status Weighting:** This is a new, critical step. Assign a higher intrinsic weight to all `CONNECTED` data points. A simple quality multiplier (`quality_w`) can be applied (e.g., `quality_w = 2.0` for `CONNECTED`, `quality_w = 1.0` for `SCAN`). This ensures that in all subsequent averaging or modeling, the high-quality connected points have more influence from the outset.
- **linkSpeed as a Quality Proxy:** For `CONNECTED` points, the `linkSpeed` serves as an excellent proxy for connection stability. A measurement with an unusually low `linkSpeed` despite a high RSSI might indicate interference. The `quality_w` for these points can be down-ranked accordingly.

Stage 2: Advanced Outlier Detection for Location Data

After initial checks, the pipeline must address the more subtle problem of spatial outliers in the location data.

Global vs. Local Spatial Outlier Detection

Spatial outliers can be categorized as either global or local, and different techniques are required to detect them.[15]

- **Global Outliers:** A global outlier is a point that is far away from all other points in the dataset for a given AP. A straightforward method for their detection is to first compute the geometric centroid of all measurement locations for an AP. Then, any point whose distance from this centroid exceeds a robust threshold (e.g., three times the median absolute deviation) can be flagged as a global outlier and removed.[15]
- **Local Outlier Factor (LOF):** A more nuanced technique is the Local Outlier Factor (LOF) algorithm, which identifies local outliers.[15] LOF works by comparing the local density of a point to the local densities of its neighbors. A point that is in a much sparser region than its neighbors will receive a high LOF

score and can be removed.

Stage 3: Filtering and Denoising RSSI Measurements

The final pre-processing stage is to address the noise inherent in the RSSI values themselves. An FCM-Kalman combination is recommended to handle the likely non-Gaussian nature of RSSI noise.

4. A Multi-Algorithm Strategy for Access Point Localization

The core of the AP localization process must employ a multi-algorithm strategy, dynamically selecting the most appropriate technique based on the maturity and quality of the data available for each AP. The distinction between CONNECTED and SCAN data is paramount here.

The Principle of Dynamic Algorithm Selection based on Data Maturity

The system employs a tiered approach. A localization attempt is only made once a minimum bootstrap sample of data has been collected (e.g., $N \geq 20$) to ensure statistical robustness. After this initial calculation, the system graduates to more sophisticated models as data maturity increases.

Tier 1: Bootstrap Localization (e.g., $20 \leq N < 50$)

The first time an AP's location is calculated is during the bootstrap phase. This occurs only after a sufficient number of measurements ($N \geq 20$) have been collected and filtered. The primary algorithm for this phase is Weighted Centroid Localization (WCL).

Weighted Centroid Localization (WCL)

- **Principle:** WCL computes the AP's location as the weighted average of the measurement coordinates.
- **Enhanced Formula:** The weighting function is now enhanced to include the data quality weight from Stage 1. The total weight w_i for each point is a product of its signal strength and its connection quality:

$$w_i = \text{quality_w}_i \times 10^{\{\text{RSSI}_i / 10\}}$$

This gives the CONNECTED points significantly more influence in the initial bootstrap estimate.

- **Limitations:** WCL remains highly susceptible to geometric bias. A "Situation Goodness" check is still required as a quality gate.

Advanced Probabilistic Models for Mature Datasets (High Data Volume: e.g., $N \geq 50$, High SDS)

For mature datasets, the system uses sophisticated probabilistic models. Here, the rich connection_metadata provides a transformative advantage.

Maximum Likelihood Estimation (MLE) and Bayesian Inference Frameworks

- **Principle:** These models seek the AP location L_{AP} that maximizes the likelihood of observing the collected set of RSSI measurements, based on a physical signal propagation model.
- **The Log-Distance Path Loss Model:** $RSSI(d) = A - 10 * n * \log_{10}(d) + X_{\sigma}$
- **Leveraging CONNECTED Data for a High-Fidelity Model:** This is the most critical improvement. The model parameters can now be set with far greater precision for CONNECTED measurements:
 - **Path-Loss Exponent (n):** For a CONNECTED point, the system can use the exact frequency (e.g., 5660 MHz) and channelWidth from the metadata to select a highly specific and accurate path-loss exponent n from a pre-computed lookup table. For a SCAN point, the system must fall back to a generic, less accurate n based on a broad frequency band guess.
 - **Measurement Noise (X_{σ}):** For a CONNECTED point, the RSSI is more stable. Therefore, the model can use a **lower standard deviation (σ)** for the noise term, reflecting higher confidence in the measurement. For a SCAN point, a **higher σ** must be used to account for the greater uncertainty of a passive scan.
- **Bayesian Inference:** In the Bayesian framework, this higher-quality modeling for CONNECTED points results in a much tighter likelihood function. When combined with the prior, this produces a posterior distribution with a smaller variance, representing a more confident and accurate final estimate.

Table 2: AP Localization Algorithm Selection Matrix (Updated)

Data Maturity Tier	Criteria	Recommended Primary Algorithm	Key Data Inputs & Models
Tier 1: Bootstrap	$20 \leq N < 50$	Weighted Centroid Localization (WCL)	This is the <i>first</i> localization attempt. Requires a "Situation Goodness" check. Uses quality_w to give preference to CONNECTED points.
Tier 2: Mature	$50 \leq N < 100$ and SDS is High	Maximum Likelihood Estimation (MLE)	Uses specific n and low σ for

			CONNECTED points; generic n and high σ for SCAN points.
Tier 3: Highly Mature	$N \geq 100$ and SDS is High	Bayesian Inference	Same as MLE, but incorporates the prior estimate and covariance for iterative refinement.

5. Statistical Fusion and Confidence Modeling

Producing a single coordinate pair from a localization algorithm is only half the task. For a system to be robust and capable of iterative improvement, it must also quantify its confidence in that estimate. This uncertainty is not merely an error metric for reporting; it is a first-class citizen in the data model, a critical input that drives the entire iterative refinement loop.

Quantifying Uncertainty: Error Covariance and Confidence Scores

- **Error Covariance Matrix:** The most comprehensive representation of uncertainty for a 2D location estimate is the 2x2 error covariance matrix P . This is a natural output of the Bayesian and Kalman Filter frameworks.
- **Confidence Score:** For easier interpretation, the covariance can be distilled into a single confidence score (0-100) based on variance, N , and SDS.

The critical takeaway is that this uncertainty information, particularly the covariance matrix, must be persisted as part of the AP's state to enable statistically sound iterative learning.

6. The Iterative Refinement and Online Learning Loop

The system architecture must include an iterative refinement loop that treats the AP's location not as a value to be calculated once, but as a state to be estimated and tracked over time using a recursive filter like the Kalman filter.

Accelerating Convergence with High-Quality Data

The distinction between CONNECTED and SCAN data directly accelerates this learning loop.

- **Refining the Kalman Filter Update:** The "measurement noise" (R_k) fed into the Kalman filter represents the uncertainty of the new location estimate generated by the localization module.

- **Impact:** When a location estimate is generated using a high proportion of CONNECTED data, the resulting model is more accurate. This leads to a smaller error covariance matrix R_k (indicating lower uncertainty). When this high-confidence measurement is fed into the Kalman filter, the Kalman Gain gives it more weight. This causes the AP's "golden record" location to converge towards the true value much more quickly and with higher final accuracy.

The Feedback Loop for Continuous Improvement

1. A new batch of M measurements arrives.
2. The system retrieves the AP's current state (location $\hat{x}_{k-1|k-1}$ and covariance $P_{k-1|k-1}$).
3. This state is used as the *prior* for the Bayesian Inference model.
4. The Localization Module runs, using the high-fidelity models for CONNECTED data, producing a new estimate (z_k) and its covariance (R_k).
5. The Kalman filter is invoked, optimally fusing the prior state with the new measurement.
6. The filter produces the new, refined state ($\hat{x}_{k|k}$, $P_{k|k}$), which is persisted.

Detecting Relocation via the Filter

The Kalman filter's **innovation** (the residual between the measurement and the prediction) remains a powerful signal for drift detection. A consistently large innovation indicates the AP has likely moved, triggering the state reset procedure.

7. Synthesis and Unified Process Flow

The entire process is a continuous pipeline with a critical feedback loop.

1. **Data Ingestion & Tiering:** Raw records are ingested, batched per BSSID, and explicitly labeled as CONNECTED or SCAN.
2. **AP Dynamic State Classification:** Run Hotspot and Relocation Detection.
3. **Pre-processing and Filtering:** Apply sanity checks and assign a *quality_w* based on connection status. Run advanced outlier detection (Global and Local).
4. **Dynamic Localization:** Select and execute the appropriate algorithm (WCL, MLE, or Bayesian), using the enhanced, high-fidelity models for CONNECTED data.
5. **State Estimation and Refinement:** Feed the new estimate and its covariance into the AP's Kalman filter to produce a new, refined state.
6. **State Persistence:** Write the new state and updated maturity metrics back to the AP database.

By implementing this complete framework—which formally distinguishes between

high- and low-quality data tiers and uses rich metadata to inform physical models—it is possible to build a truly robust, accurate, and fast-converging Wi-Fi location database from dynamic, real-world crowdsourced data.

8. Future Directions: Incorporating Wi-Fi RTT and Advanced ML

While the proposed framework provides a robust foundation based on RSSI, the field of Wi-Fi positioning is continuously evolving. The system architecture should be designed with forward compatibility in mind.

- **Wi-Fi RTT (IEEE 802.11mc):** The Fine Time Measurement (FTM) protocol, increasingly supported by Android devices, allows for direct distance measurement between a device and an AP by measuring the signal's round-trip time.[6, 50] This is a paradigm shift, as RTT provides a direct distance estimate that is far more stable and less susceptible to environmental effects than RSSI.[50, 51] The localization pipeline should be designed to ingest and utilize RTT measurements when available (e.g., when `is80211mcResponder` is true). In the MLE/Bayesian models, an RTT measurement can be used to directly inform the distance d , bypassing the noisy RSSI-to-distance conversion and dramatically improving the accuracy and confidence of the likelihood model. The fusion of RTT and RSSI is a powerful technique for next-generation systems.[52]
- **Deep Learning (DL):** Deep Neural Networks offer a powerful alternative to the explicit probabilistic models.[14, 48] A DNN could be trained to learn the complex, non-linear mapping directly from a raw RSSI vector (and other sensor data) to a location. Such models have shown great promise in handling challenges like device heterogeneity and dynamic environmental changes without explicit modeling.[53, 54] In a future iteration of the system, the Localization Module could be augmented or replaced with a trained DL model. The system's data pipeline would serve as an excellent source of training data for such a model.